

# Long-Term Outcomes of Dental Implants Placed in Fibula-free Flaps Used for Reconstruction of Maxillo-Mandibular Defects



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## KEYWORDS

- Dental implants • Fibula • Free flap • Mandibular reconstruction • Microvascular
- Revascularized-free flaps

## KEY POINTS

- The efficacy and prognosis of dental implants placed in free bone flaps have been extensively studied, with the fibula free flap being an ideal option for mandibular reconstruction with dental rehabilitation.
- Patients reconstructed with free fibula flap (FFF) and implant-supported overdentures showed statistically significant improvements in quality of life (QOL) when compared with no prosthesis or non-implant-supported prosthesis in post ablative defects.
- Dental implants can be placed in free fibula flaps using immediate, secondary, or jaw in a day technique. These techniques should be individualized based on the patient's diagnosis, history, prognosis, motivation, needs and wishes.
- Success rates of dental implants placed in nonirradiated fibulas are comparable with those placed in native bone.
- Implants inserted into the irradiated maxilla have a lower survival rate than implants inserted into the irradiated mandible and irradiated FFF implants have a lower survival rate than native maxilla or mandible implants.

## INTRODUCTION

When bone integrity is compromised due to trauma, infection, large jaw cysts, or ablative oral cancer surgery, jaw reconstruction is frequently required.<sup>1</sup> Reconstruction of the maxilla or mandible using a fibula free flap is the recognized standard of care when surgical resection involves segmental resection.<sup>2</sup> An increasingly popular treatment strategy for patients with head and neck cancer is the use of dental endosseous implants and overdentures

as part of their oral and dental rehabilitation. There are several advantages to implant anchorage compared to traditionally secured prostheses, but most importantly is the significant improvement in the reported quality of life of patients.<sup>3</sup> The first reports of dental prosthetic rehabilitation of the mandible using free tissue transfer and endosseous implants dates back to 1989.<sup>4,5</sup> The efficacy and prognosis of implants placed in FFF has been extensively studied. The ease of harvest, availability of long span of bone, segmental periosteal blood

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supply, length of vascular pedicle, pliability of skin paddle, reliable vascular anatomy, and ability to use a 2-team approach make the fibula free flap an ideal option for mandibular reconstruction with dental rehabilitation.

## SUCCESS RATES

Implants placed in native bone have similar success rates to those placed in non-irradiated fibulas. After one to 5 years of placement, the average overall survival rate of implants placed in FFF is 93.5%, with a range of 83.3% to 97%.<sup>6–16</sup> These implants have an average 10-year survival rate of 80% and a reported 20-year survival rate of 69%.<sup>12</sup>

Kolokythas and colleagues conducted a meta-analysis with data obtained from 242 patients and a total of 848 dental implants placed in free fibula flaps secondarily. The estimated proportion of successful implants placed in fibula flaps used to reconstruct the maxillomandibular complex was 0.94 or 94% (95% CI [confidence interval] = 0.91–0.96)]. The annual implant failure rate was 0.02 with a 95% CI = 0.01 to 0.03.<sup>17</sup>

Liu and colleagues conducted a meta-analysis to compare the survival rate and complications between primary and secondary implantation specifically associated with vascularized fibula transplantation for reconstruction of jaw defects. 7 studies were involved with 186 patients. The survival rate of the primary implantation was 93.3%. The survival rate of the secondary implantation was 93.4%. The meta-analysis further proved that there was no significant difference in the survival rate between primary implantation and secondary implantation, OR = 0.813 (95% CI 0.383–1.725,  $P = .589 > 0.05$ )<sup>18</sup>

Ch'ng et al investigated the rate of implant loss in patients with head and neck cancer who had implant placement in fibulas and native jaws before and after radiation treatment. They discovered that, while not statistically significant, there was a slightly higher rate of loss in fibulas (8.6%) than in native mandibles (2.6%) and maxillas (2.2%).<sup>10</sup> Furthermore, there was no statistically significant variation in implant loss between patients who underwent adjuvant radiation therapy and those who did not. When comparing patients who had a fibula and primary implant placement to those who had not, Sandoval and colleagues specifically studied the effects of adjuvant radiation therapy in patients reconstructed with a fibula and primary implant placement compared with fibulas without implant placement and found no significant difference in adverse outcomes.<sup>19</sup>

Radiation and pathology did not significantly affect implant survival statistically.<sup>20</sup> Implants inserted into non-irradiated fibulas have success rates that are similar to those of implants inserted into native bone. After one to 5 years of placement, the average overall survival rate of implants placed in FFF is 93.5%, with a range of 83.3% to 97%.<sup>7,15</sup> These implants have an average 10-year survival rate of 80% (range: 78%–83%), and a reported 20-year survival rate of 69%.<sup>15</sup> Maxillofacial subunit reconstruction using an implant-supported prosthesis and FFF has a high implant survival rate, minimal crestal bone loss (<1 mm), and minimal complication rates, making it durable over the long run. Implants placed in FFF exhibit a higher overall implant loss<sup>10</sup> and peri-implant bone loss in comparison to native bone, especially when placed in the maxilla.<sup>10,19</sup>

## VIRTUAL SURGICAL PLANNING

Virtual surgical planning is a major factor in both primary and secondary implant placement. It involves computer-aided surgical planning and the creation of surgical guides using computer-aided design and manufacturing (CAD/CAM) technology. Virtual surgical planning (VSP) enables accurate evaluation of anatomy, creation of a patient-specific plan for FFF and implant placement, and implementation of this plan using custom-fabricated cutting and implant placement guides.<sup>4</sup> By using this technology, endosseous implant placement can be planned while avoiding the reconstruction plate and screws (Fig. 1). The precise cutting of the fibula segments in patient-specific reconstruction plates, as well as the planning of osteotomies in the defect site, are made possible by prefabricated fibula cutting and implant placement guides made using VSP (Fig. 2). Moreover, it permits the optimal positioning



**Fig. 1.** Panorex reconfigured from a cone-beam computed tomogram showing mandibular segmental reconstruction with implants and prosthesis in place. (Data from Salam O. Salman, Rui P. Fernandes, Sundeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1–2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)



**Fig. 2.** Fibula osteotomy guide with integrated implant drill guide. (Data from Salam O. Salman, Rui P. Fernandes, Sundeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1-2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)

of endosseous implants in the fibula bone, with pre-defined angulation and position that facilitates prosthetic rehabilitation, and it prevents interference between the implant and the transosseous screws that secure the FFF.<sup>19-21</sup>

This results in increased rates of tertiary rehabilitation and more efficient fabrication and placement of a final prosthesis. Reduced operating times, fewer procedures, increased utilization of implants, shorter prosthesis delivery times, higher rates of tertiary prosthetic rehabilitation, and enhanced patient satisfaction and quality of life make up for the higher cost of VSP.<sup>4</sup> (Fig. 3)

#### IMMEDIATE, SECONDARY, AND JAW IN A DAY

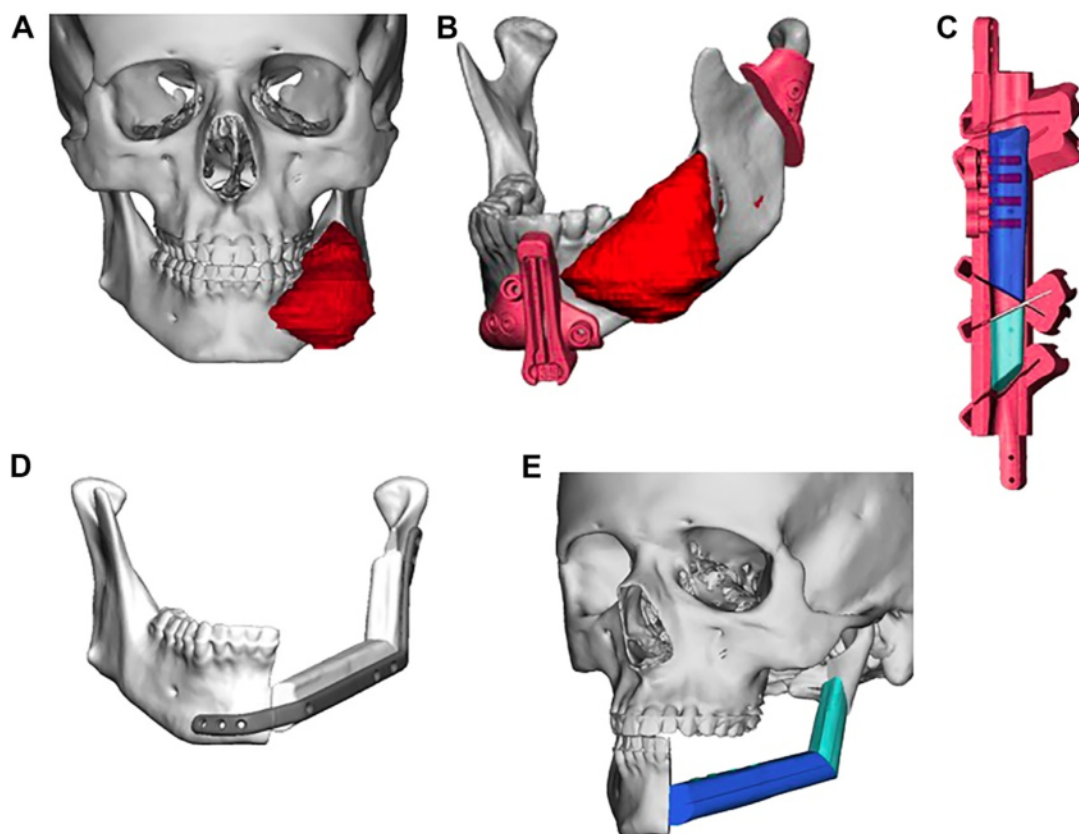
A number of factors make the post-ablative patient an unsuitable candidate for conventional (non-

implant-supported) dentures: low bone density, poor soft tissue quantity and quality, complications related to radiation therapy, such as soft tissue loss and xerostomia leading to ulceration from acrylic rubbing against the mucosa, poor denture retention, and poor suction.

The placement of dental implants in free bone flaps can be done using 1 of 3 methods: "Jaw in a day," primary, or one-stage, and secondary, or 2-stage. Primary or one-stage implant placement in FFF can be safely carried out without vascular compromise to the flap. It adds extra time to the total surgical period, and overall FFF warm ischemia time is less than 4 hours. This procedure improves functional outcome and is cost-effective because it decreases overall reconstruction time by reducing the number of OR visits. Immediate implant placement has several benefits, such as easier access to the fibula bone, fewer surgical procedures, reduced cost, early oral rehabilitation, and a quicker return to oral nutrition and prosthesis use. It also gives implants time to osseointegrate before beginning radiation therapy and may lower the risk of implant loss. Implant placement errors, radiation therapy interference, changes to the local anatomy, and the potential for tumor recurrence at the implant placement site that hinders surveillance are among the drawbacks of this technique. Primary implant placement following the excision of a malignant tumor is not advised due to the possibility of recurrence.<sup>22</sup>

The placement of secondary implants is done in 2 steps. The bone defect is reconstructed in the first stage using a reconstruction plate and FFF, after which it is left to heal for 4 to 6 months. The optimal location and angulation for implant placement are determined in the second stage by the surgeon with the assistance of VSP technology. The secondary implant placement is postponed for a year following the conclusion of therapy in malignant tumor defects treated with FFF and adjuvant radiation therapy. In second-stage surgery, reconstructive hardware is removed if it gets in the way of implant placement. Vestibuloplasty, flap skin paddle debulking, and surrounding implant grafting can be completed concurrently with implant placement or during implant uncovering.<sup>13</sup> (Fig. 4) Secondary implant placement has several benefits, such as identifying patients who are motivated to pursue additional prosthetic rehabilitation, a shorter initial surgical procedure, developing an optimal implant placement guide and prosthodontic plan, and providing time to rule out local tumor recurrence.<sup>23-25</sup> The drawbacks include higher costs, more OR visits, a longer time for oral rehabilitation.



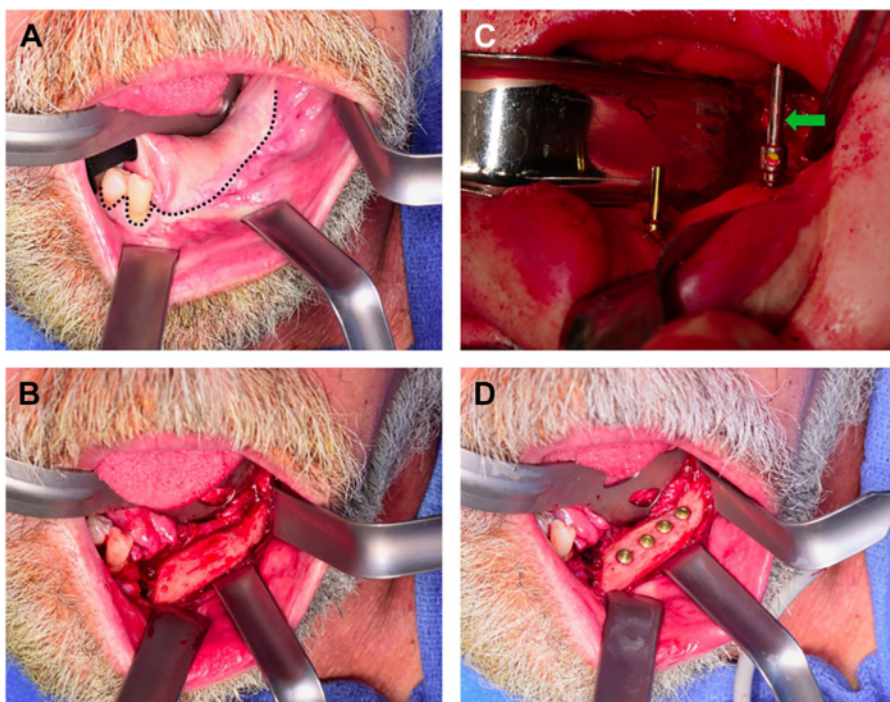


**Fig. 3.** Virtual surgical planning for post-tumor ablation mandibular reconstruction. (A) A 3D computer-generated model displaying the mandibular mass on the left side of the face. (B) Mandible with a customized cutting guide and depicting tumor margins. (C) Patient-specific guide for fibula osteotomy. (D) The reconstructed mandible demonstrates how the fibula segments and reconstruction plate should be positioned in relation to the natural mandible and dentition. (E) The reconstructed mandible featuring the patient-specific reconstruction plate and the fibula graft. (Data from Salam O. Salman, Rui P. Fernandes, Sundeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1-2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)

With today's virtual technology, the fibula free flap, guided implants as well as the dental prostheses can all be placed in a single surgical procedure (Fig. 5). Despite the limited experience of Jaw in a Day with malignant disease, preliminary reports indicate that it can be successfully used for patients who require adjuvant radiation therapy following surgery. It is likely that radiation therapy has detrimental effects on bone that take several weeks to manifest. Since radiation usually starts 6 weeks after surgery, a large portion of the osseointegration process has already taken place before higher radiation doses are accumulated.

The primary implant placement strategy of placing implants in nonradiated bone has been supported by prior research. Implant placement prior to radiation therapy appears to have no effect on the rate of successful integration and may even slightly increase the survival rate.<sup>26-29</sup>

The introduction of CAD-CAM and the high success rates linked to this method have spurred a trend toward rapid prosthesis fabrication and implantation, as several case studies and reports have shown.<sup>30-34</sup> Chiapasco and Gatti were the first to demonstrate the immediate prosthetic loading of implants. In their study, 2 patients with severely atrophied, edentulous maxilla were treated with fibula free flap reconstruction followed by the placement of the prosthesis and implants 3 months later.<sup>34</sup> Every implant was found to have survived its owner. Similarly, Okay and colleagues examined 28 patients and found that, 3 months after free flap reconstruction, patients who had immediate loading of a provisional prosthesis had an implant success rate of 89.3%.<sup>35</sup> These findings opened the door for single-stage total maxillomandibular reconstruction by indicating that there was potential for immediate functional loading of implants (Fig. 6).



**Fig. 4.** Implant placement in the free fibula flap. Mandible intraoral photograph (A) displays incision marks. (B) In order to keep the segment from becoming devascularized, a small amount of subperiosteal dissection is done over the fibula following the incision. Using an implant placement guide and paralleling pins, implants are positioned in a standard manner, ensuring that their angulation is accurate. (D) Cover screws hold implants in place. (Data from Salam O. Salman, Rui P. Fernandes, Sundeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1-2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)

## TECHNIQUES

At the head, the fibula bone has a triangular shape; in the middle, it becomes more quadrilateral, and at the malleolus, it becomes oval or irregular. In most cases of reconstructed defects, the distal fibula



**Fig. 5.** Pedicled fibula after osteotomies, plating and placement of implants with screw-retained prosthesis. (Data from Salam O. Salman, Rui P. Fernandes, Sundeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1-2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)

bone—which is typically quadrilateral in shape—is used to enable a longer vascular pedicle length.

The ideal position for the fibula bone is one that avoids the screws from the reconstruction plate and permits the angulation of the implant placement. Ideally, segments of the osteotomized fibula should not be smaller than 2 cm.<sup>36,37</sup>

Although fibulas positioned at the inferior border produce aesthetically pleasing results, the prosthesis must have a significantly taller structure due to the resulting lack of vertical "alveolar" height, this can result in higher loading forces on the implants.<sup>38</sup> One of the ways to prevent this is by placing the fibula segment 5 to 10 mm above the mandibular inferior border. Another important consideration is the space needed to place the implants, take dental impressions, allow mastication, and deliver a prosthesis a minimum of 15 mm of mouth opening is needed at the incisal edge of the anterior teeth. There should be a 10 to 15 mm space between the superior edge of the fibula and the occlusal edge of the opposing dentition to enable the development of the implant-supported framework of the prosthesis.<sup>26</sup> The placement of implants in native jaw bone follows the same guidelines as the





**Fig. 6.** (A, B) Intraoral images after completion of mandibular reconstruction with prosthesis in place. (Data from Salam O. Salman, Rui P. Fernandes, Sundeeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1-2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)

placement of implants in fibula bone, with the exception of the high bone density fibula, where bone fracture after implant drilling must be avoided by tapping the implant osteotomy site.

Endosseous implant placement requires the fibula bone to be at least 10 mm in height and 5 mm wide.<sup>21,39</sup> Bicortical implant placement is crucial in FFF due to the absence of cancellous bone in order to achieve stability and higher removal torque values.<sup>32</sup> It is recommended to plan for a maximum of 2 implants per 2-cm fibula segment. To preserve the maximum amount of bone vascularity, segments should be stabilized using a reconstruction plate and 1 or 2 screws.<sup>26</sup>

In order to stabilize longer fibula segments, prevent micromovement at the osteotomy site, and minimize rotational forces on the segments, a minimum of 2 monocortical screws should be used. The recommended minimum spacing between 2 implants is 3 mm as in the native mandible. Additionally, the implant should be at least 0.5 mm from the cortical plate of the lingual or buccal fibula bone, and at least 3 mm from the osteotomy site.

For the purpose of prosthetic rehabilitation, implants ought to be oriented parallel to each other and with the proper angulation. Angled abutments allow for a maximum of 15° of divergence during prosthesis fabrication. Implant insertion torque should be set at a minimum of 20 Ncm and a maximum of 45 Ncm, respectively.<sup>40</sup> A surgical guide using bone, mucosa, or tooth support can be used to direct the implantation of endosseous implants in the bone. Implants can be precisely positioned, angled, and spaced with a 1 mm margin of error when implant placement guides are used. A screw-retained prosthesis can be provided immediately if the implant insertion torque is 35 Ncm or higher. A 2-stage protocol is employed if the insertion torque is less than 35 Ncm.<sup>41–45</sup>

In order to achieve a consistent outcome, accurate preoperative planning is essential. A computed tomography (CT) scan of the fibulas and jaws with

1 mm cut is first obtained. Using an intraoral optical scan, a digital file that can be used as a template for making a temporary dental prosthesis and as a reference for implant positioning can be created. The stereolithography digital format is the most widely used. When a patient's preoperative dentition is intact, a temporary prosthesis can be made by 3D printing the teeth that were cloned using an optical scanner.<sup>45,46</sup>

For the virtual planning session, the intraoral scan digital file must be uploaded to the VSP planner in order for it to match the CT data. Dental implant placement, fibula orientation, mandible resection location, and plate/screw design comprise the 4 components planned using VSP. Optimal fibula positioning should be the foundation for mandible osteotomies in addition to margins that are clinically sound. Sometimes it is necessary to extend the resection margins in order to avoid short fibula segments, which might have a less robust blood supply.<sup>7,45–47</sup>

## TYPE OF IMPLANTS AND PROSTHESIS

Implant survival is not influenced by the brand of the implant.<sup>48</sup> The majority of implant systems that are sold commercially are equally suitable for FFF dental restoration.<sup>39</sup> When compared to machined implants, implants with treated surfaces have comparatively fewer failures. This is probably because rough-surface implants have a larger surface area and accelerate osseointegration. When placed in irradiated fields, short implants have higher failure rates than standard-size implants (10 mm or longer).<sup>49</sup> In light of these findings, FFF reconstruction should employ a surface-treated implant with a minimum width of 3 mm and a minimum length of 10 mm.<sup>50</sup>

Bar connectors are preferable to ball-and-socket connectors because they minimize implant micro-movements and resolve issues with misaligned or incorrectly positioned implants. A removable

implant-supported prosthesis with ball-and-socket connectors has a higher failure rate.<sup>51</sup>

Poor intermaxillary relationships, tumor recurrence, implant loss, microstomia, and a lack of patient cooperation are among the factors that contribute to the failure of prosthetic rehabilitation. A poor emergence profile, increased loading forces, and stress on the implants are caused by implants with poor vectors, misaligned arches, long abutments, and increased distances between the fibula and opposing occlusal surface. These factors may compromise the long-term survival of the implants.<sup>52</sup> A minimum of 4 implants should be positioned per arch for edentulous patients receiving treatment with implant-supported overdentures. In 2016, Kumar and colleagues<sup>53</sup> found no difference in the quality of life (QOL) between patients treated with overdentures based on 2 versus 4 implants, despite the fact that patients treated with the 2-implant system had higher marginal bone loss.

Ideally 4 implants with bar attachments are used to support the removable implant-supported overdenture. The prosthesis survival rate for patients who have undergone rehabilitation is high—more than 98%.<sup>54</sup> Patients who have FFF and overdentures supported by implants fare better than those who do not. Patients reconstructed with FFF and implant-supported overdentures showed statistically significant improvements in QOL when compared with no prosthesis or non-implant-supported prosthesis in post ablative defects. For these patients, there are restorative options such as hybrid prosthesis, overdentures, and implant-supported cement or screw-retained bridges.<sup>53</sup>

Implant-supported obturators can be utilized by maxillectomy patients to seal oronasal communications created by the defect.

In order to minimize the risk of pressure necrosis and soft tissue ulceration, the best option would be a fixed hybrid screw-retained prosthesis that does not rely on the mucosa.<sup>55</sup> Cement-retained prostheses and acrylic restorations should be avoided as they cause granulation tissue formation around the implant and bone interface (Fig. 7).

## SOFT-TISSUE MANAGEMENT

In order to reconstruct the soft-tissue defect, the FFF harvest is frequently combined with a skin paddle. If radiation therapy is necessary after ablation and reconstruction, implant placement typically occurs 6 months after the completion of radiation therapy.<sup>49</sup>

Various techniques have been reported to result in attached tissue surrounding implants upon exposure; these encompass debulking the flap skin paddle, placement of split or full-thickness skin grafts,



**Fig. 7.** Peri-implantitis around mandibular implants resulting in granulation tissue and persistent inflammation. (Data from Salam O. Salman, Rui P. Fernandes, Sundeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1-2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)

as well as palatal mucosa grafts. The skin paddle separates the oral cavity from the neck, sinuses, and nasal cavity, providing the required bulk and integrity for reconstruction. In order to prevent peri-implantitis, proper soft-tissue management involves thinning out any bulky or redundant tissue from the skin paddle, creating vestibules for the lips, labial, and buccal regions, allowing unrestricted tongue mobility, and enhancing cosmesis.<sup>4</sup>

Debulking the skin paddle and using split-thickness skin grafts to provide keratinized tissue around implants have demonstrated encouraging outcomes, with no notable complications with hyperplastic granulation tissue forming at the implant and skin interface.<sup>10</sup>

Of all the options, the palatal mucosa graft has proven to be the best option because it provides keratinized attached mucosa that forms a seal around the implants, permits proper hygiene, and inhibits granulation tissue formation and bone loss surrounding the implants.<sup>38,56</sup>

During the healing phase following implant exposure, the use of skin and mucosal grafts in conjunction with a custom-fabricated compressive splint lined with tissue conditioner can inhibit the formation of granulation tissue around the abutments.<sup>57</sup> These splints can be used for vestibuloplasty concurrently, and they should be left in place for 6 to 8 weeks in order to allow the graft to firmly adapt to the underlying tissue and form a vestibule.

## COMPLICATIONS

Peri-implantitis, which is also the most frequent cause of implant loss in FFF, is one of the most



frequent issues related to implant placement in this population. There have been reports of osteomyelitis, fibula devascularization, fibula fracture, osteoradionecrosis (ORN), and tumor recurrence at the implant placement site.<sup>23</sup> All instances of tumor recurrence at implant locations happened in implants that were mostly placed primarily.<sup>58</sup> Recurrence at primary implant sites may result from peri-implantitis, long-term alcohol and tobacco use, interference with radiation dosage from metallic implants, and a general increased susceptibility to chronically irritated oral mucosa. To rule out oral squamous cell carcinoma, chronic peri-implantitis should be closely monitored, treated, and biopsied.<sup>59</sup>

There are a number of drawbacks to using a fibula flap along with dental rehabilitation, such as: insufficient bone height, interference of the skin paddle, poor skin graft take at the donor site, immobilization of the leg, and possible vascular compromise of the foot.<sup>60</sup> The outcome of a split-thickness skin graft at the FFF donor site may also be inconsistent.

The absence of vertical height in the FFF bone is the main drawback. The vertical height of an adult dentate mandible is approximately half that of the fibula bone, which is 13 to 15 mm. The reconstruction plate and fibula can be positioned 5 to 10 mm higher to get around this problem.<sup>50</sup> (Fig. 8) This method would result in a minor cosmetic defect at the inferior border, but it can be fixed by placing a deepithelialized vascularized fibula skin paddle over the reconstruction plate to compensate for the loss of soft tissue and higher placement of the fibula bone. Adding nonvascularized corticocancellous autograft to rebuild the alveolus, double



**Fig. 8.** Position of the fibula segment at a higher level while maintaining the reconstruction plate at the mandible's inferior border. (Data from Salam O. Salman, Rui P. Fernandes, Sundeep R. Rawal, Immediate Reconstruction and Dental Rehabilitation of Segmental Mandibular Defects: Description of a Novel Technique, *Journal of Oral and Maxillofacial Surgery*, 75 (10), 2017, 2270.e1-2270.e8, <https://doi.org/10.1016/j.joms.2017.06.019>.)

barreling and vertical distraction osteogenesis are additional strategies to overcome this.<sup>61-64</sup> Placing immediate, well-positioned implants in a double-barreled fibula can be difficult. Vertical fibula distraction is unpredictable and can result in lingual tipping of the superior segment.<sup>42</sup>

## EFFECTS OF RADIATION

The timing of radiation therapy before or after implant placement does not have an impact on implant survival, according to more recent data.<sup>9,23</sup> The development of intensity-modulated radiation therapy, which has been demonstrated to have statistically higher implant survival than conventional radiation therapy, may help explain this finding.

While the exact timing of implant placement following radiation therapy does not affect overall implant survival, most studies advise delaying implant placement for at least 6 to 12 months.<sup>39,62</sup> The radiation doses to nearby tissues are altered by metallic objects in the radiation field.<sup>65</sup>

Primary implant placement prior to radiation therapy may result in dose escalation in the bone in front of the implant, shielding behind the implant, and backscattering.<sup>58,66</sup> Greater implant sizes result in more localized x-ray or gamma-ray back scatter, which raises local radiation doses and increases the risk of ORN. The dose increases by 15% at the interface between the implant and bone due to backscatter, but it drops to almost nothing within 1 to 2 mm of the interface.<sup>67</sup>

The effects of backscatter are mitigated by multi-vectored radiation therapy. Because of this effect, it is advisable to place the smallest clinically feasible implants in order to prevent higher radiation doses and increased backscatter at the bone implant interface. With an average overall implant survival of 91.33% over 1 to 5 years<sup>10,16,48</sup> and a long-term implant survival rate of 78% to 85% at 15 years, implants placed in the irradiated field have a high success rate. Implants inserted into the irradiated maxilla have a lower survival rate than implants inserted into the irradiated mandible<sup>49</sup> and irradiated FFF implants have a lower survival rate than native maxilla or mandible implants.<sup>10</sup>

Additionally, there is a noticeably increased risk of implant loss in previously irradiated FFF that has already developed ORN.<sup>10,25,68</sup>

According to a systematic review conducted in 2016 by Filho and colleagues,<sup>69</sup> there is no significant correlation between the time of implant placement following radiation therapy and the radiation dose in terms of implant loss. Hyperbaric oxygen therapy is controversial when used during the peri-implant placement period. According to available data, its use does not appear to be



associated with a higher risk of implant failure and does not appear to improve implant survival.<sup>10,42,68</sup> It also does not appear to prevent ORN after implant placement<sup>12,70</sup> In both irradiated and non-irradiated fibulas, the majority of implant failures occur within the first 6 to 12 months following implantation.<sup>25,71,72</sup>

## SUMMARY

The use of FFF and implant-supported prostheses for maxillary and mandibular reconstruction is a reliable and long-lasting method to give patients integrity, function, and form while enhancing their quality of life. Caution should be exercised when performing secondary implant reconstruction in patients who have previously undergone radiation therapy following the placement of FFF. To maintain the integrity of the implants that have been placed, soft-tissue management should be carried out as needed. A fixed hybrid metal ceramic screw-retained prosthesis is the ideal tool for prosthodontic rehabilitation. The preservation and success of the reconstruction depend on a determined patient, strict hygiene, and ongoing follow-up.

## CLINICS CARE POINTS

- The efficacy and prognosis of dental implants placed in free bone flaps have been extensively studied, with the fibula free flap being an ideal option for mandibular reconstruction with dental rehabilitation.
- Patients reconstructed with FFF and implant-supported overdentures showed statistically significant improvements in QOL when compared with no prosthesis or non-implant-supported prosthesis in post ablative defects.
- Dental implants can be placed in free fibula flaps using immediate, secondary, or jaw in a day technique. These techniques should be individualized based on the patient's diagnosis, history, prognosis, motivation, needs and wishes.
- Success rates of dental implants placed in nonirradiated fibulas are comparable with those placed in native bone.
- Implants inserted into the irradiated maxilla have a lower survival rate than implants inserted into the irradiated mandible and irradiated FFF implants have a lower survival rate than native maxilla or mandible implants.

## DISCLOSURE

The authors have nothing to disclose.

## REFERENCES

1. Wong HM. Oral complications and management strategies for patients undergoing cancer therapy. *Sci World J* 2014;2014:1–14. <https://doi.org/10.1155/2014/581795>.
2. Shah JP, Gil Z. Current concepts in management of oral cancer – Surgery. *Oral Oncol* 2009;45(4–5):394–401. <https://doi.org/10.1016/j.oraloncology.2008.05.017>.
3. Lavery DP, Kelly RD, Addison O. Survival of dental implants placed in autogenous bone grafts and bone flaps in head and neck oncology patients: a systematic review. *Int. J. Implant Dent.* 2018;4(1). <https://doi.org/10.1186/s40729-018-0131-9>.
4. Patel S, Kim DD, Ghali GE. Maxillofacial Reconstruction Using Vascularized Fibula Free Flaps and Endosseous Implants. *Oral Maxillofac Surg Clin* 2019; 31(2):259–84.
5. Urken ML, Buchbinder D, Weinberg H, et al. Primary placement of osseointegrated implants in microvascular mandibular reconstruction. *Otolaryngol Head Neck Surg* 1989;101(1):56–73.
6. Garrett N, Roumanas ED, Blackwell KE, et al. Efficacy of conventional and implant-supported mandibular resection prosthesis: study overview and treatment outcomes. *J Prosthet Dent* 2006;96(1):13–24.
7. Gbara A, Darwich K, Li L, et al. Long-term results of jaw reconstruction with microsurgical fibula grafts and dental implants. *J Oral Maxillofac Surg* 2007; 65(5):1005–9.
8. Chiapasco M, Colletti G, Romeo E, et al. Long-term results of mandibular reconstruction with autogenous bone grafts and oral implants after tumor resection. *Clin Oral Implants Res* 2008;19(10):1074–80.
9. Jackson RS, Price DL, Arce K, et al. Evaluation of clinical outcomes of osseointegrated dental implantation of fibula free flaps for mandibular reconstruction. *JAMA Facial Plast Surg* 2016;18(3):201–6.
10. Ch'ng S, Skoracki RJ, Selber JC, et al. Osseointegrated implant-based dental rehabilitation in head and neck reconstruction patients. *Head Neck* 2016;38(Suppl 1):E321–7.
11. Smolka K, Kraehenbuehl M, Eggensperger N, et al. Fibula free flap reconstruction of the mandible in cancer patients: evaluation of a combined surgical and prosthodontic treatment concept. *Oral Oncol* 2008;44(6):571–81.
12. Teoh KH, Huryn JM, Patel S, et al. Implant prosthodontic rehabilitation of fibula free-flap reconstructed mandibles: a Memorial Sloan-Kettering Cancer Center review of prognostic factors and implant outcomes. *Int J Oral Maxillofac Implants* 2005;20(5): 738–46.

13. Sozzi D, Novelli G, Silva R, et al. Implant rehabilitation in fibula-free flap reconstruction: a retrospective study of cases at 1-18 years following surgery. *J Cranio-Maxillo-Fac Surg* 2017;45(10):1655-61.
14. Chiapasco M, Romeo E, Coggiola A, et al. Long-term outcome of dental implants placed in revascularized fibula free flaps used for the reconstruction of maxillo-mandibular defects due to extreme atrophy. *Clin Oral Implants Res* 2011;22(1):83-91.
15. Fang W, Liu Y, Ma Q, et al. Long-term results of mandibular reconstruction of continuity defects with fibula free flap and implant-borne dental rehabilitation. *Int J Oral Maxillofac Implants* 2015;30(1):169-78.
16. Hakim SG, Kimmerle H, Trenkle T, et al. Masticatory rehabilitation following upper and lower jaw reconstruction using vascularised free fibula flap and endosseal implants—19 years of experience with a comprehensive concept. *Clin Oral Invest* 2015;19(2):525-34.
17. Gangwani P, Almana M, Barmak B, et al. What is the success of implants placed in fibula flap? a systematic review and meta-analysis. *J Oral Maxillofac Res* 2022;13(1):e3.
18. Liu YB, Wu D, Wang JY, et al. Meta-analysis of the survival rate and postoperative infection rate of primary and secondary implants after vascularized fibula transplantation for reconstruction of jaw defects. *Int. J. Implant Dent.* 2023;9(1):51.
19. Sandoval ML, Rosen EB, Robert AJ, et al. Immediate dental implants in fibula free flaps to reconstruct the mandible: a pilot study of the short-term effects on radiotherapy for patients with head and neck cancer. *Clin Implant Dent Relat Res* 2020;22(01):91-5.
20. Rana M, Chin SJ, Muecke T, et al. Increasing the accuracy of mandibular reconstruction with free fibula flaps using functionalized selective laser-melted patient-specific implants: a retrospective multicenter analysis. *J Cranio-Maxillo-Fac Surg* 2017;45(8):1212-9.
21. Cebrian-Carretero JL, Guinales-Diaz de Cevallos J, Sobrino JA, et al. Predictable dental rehabilitation in maxillomandibular reconstruction with free flaps. The role of implant guided surgery. *Med Oral Patol Oral Cir Bucal* 2014;19(6):e605-11.
22. Alzahrani BK, Fatani B, Alshalawi HS, et al. Survival of dental implants placed in iliac, fibula, and radial forearm flaps: a comprehensive review. *Cureus* 2023;15(10):e48031.
23. Raoul G, Ruhin B, Briki S, et al. Microsurgical reconstruction of the jaw with fibular grafts and implants. *J Craniofac Surg* 2009;20(6):2105-17.
24. Roumanas ED. Reconstructed mandibular defects: fibula free flaps and osseointegrated implants. *Plast Reconstr Surg* 1997;99(2):356-65.
25. Burgess M, Leung M, Chellapah A, et al. Osseointegrated implants into a variety of composite free flaps: a comparative analysis. *Head Neck* 2017;39(3):443-7.
26. Chana JS, Chang YM, Wei FC, et al. Segmental mandibulectomy and immediate free fibula osteoseptocutaneous flap reconstruction with endosteal implants: an ideal treatment method for mandibular ameloblastoma. *Plast Reconstr Surg* 2004;113(1):80-7.
27. Chang YM, Coskunfirat OK, Wei FC, et al. Maxillary reconstruction with a fibula osteoseptocutaneous free flap and simultaneous insertion of osseointegrated dental implants. *Plast Reconstr Surg* 2004;113(4):1140-5.
28. Schepers RH, Raghoobar GM, Vissink A, et al. Accuracy of fibula reconstruction using patient-specific CAD/CAM reconstruction plates and dental implants: a new modality for functional reconstruction of mandibular defects. *J Cranio-Maxillo-Fac Surg* 2015;43(5):649-57.
29. Nooh N. Dental implant survival in irradiated oral cancer patients: a systematic review of the literature. *Int J Oral Maxillofac Implants* 2013;28(05):1233-42.
30. Freudlsperger C, Bodem JP, Engel E, et al. Mandibular reconstruction with a prefabricated free vascularized fibula and implant-supported prosthesis based on fully three-dimensional virtual planning. *J Craniofac Surg* 2014;25:980-2.
31. Pauchet D, Pigot JL, Chabolle F, et al. Prefabricated fibula free flap with dental implants for mandibular reconstruction. *Eur Ann Otorhinolaryngol Head Neck Dis* 2018;135:279-82.
32. Rohner D, Bucher P, Kunz C, et al. Treatment of severe atrophy of the maxilla with the prefabricated free vascularized fibula flap. *Clin Oral Implants Res* 2002;13:44-52.
33. Schepers RH, Raghoobar GM, Vissink A, et al. Fully 3-dimensional digitally planned reconstruction of a mandible with a free vascularized fibula and immediate placement of an implant-supported prosthetic construction. *Head Neck* 2013;35:E109-14.
34. Chiapasco M, Gatti C. Immediate loading of dental implants placed in revascularized fibula free flaps: a clinical report on 2 consecutive patients. *Int J Oral Maxillofac Implants* 2004;19:906-12.
35. Okay DJ, Buchbinder D, Urken M, et al. Computer-assisted implant rehabilitation of maxillomandibular defects reconstructed with vascularized bone free flaps. *JAMA Otolaryngol Head Neck Surg* 2013;139:371-81.
36. Hidalgo DA. Fibula free flap: a new method of mandible reconstruction. *Plast Reconstr Surg* 1989;84(1):71-9.
37. Bahr W. Blood supply of small fibula segments: an experimental study on human cadavers. *J Cranio-Maxillo-Fac Surg* 1998;26(3):148-52.
38. Shen YF, Rodriguez ED, Wei FC, et al. Aesthetic and functional mandibular reconstruction with immediate



- dental implants in a free fibular flap and a low-profile reconstruction plate: five-year follow-up. *Ann Plast Surg* 2015;74(4):442–6.
39. Carbiner R, Jerjes W, Shakib K, et al. Analysis of the compatibility of dental implant systems in fibula free flap reconstruction. *Head Neck Oncol* 2012;4:37.
  40. Maluf PSZ, Ching AW, Angeletti P, et al. Insertion torque of dental implants after microvascular fibular grafting. *Br J Oral Maxillofac Surg* 2015;53(7):647–9.
  41. Bidez MW, Misch CE. Force transfer in implant dentistry: basic concepts and principles. *J Oral Implantol* 1992;18(3):264–74.
  42. Chang YM, Santamaria E, Wei FC, et al. Primary insertion of osseointegrated dental implants into fibula osteoseptocutaneous free flap for mandible reconstruction. *Plast Reconstr Surg* 1998;102(3):680–8.
  43. Meloni SM, Tallarico M, De Riu G, et al. Guided implant surgery after free-flap reconstruction: Four-year results from a prospective clinical trial. *J Cranio-Maxillo-Fac Surg* 2015;43(8):1348–55.
  44. Gherlone E, Capparé P, Vinci R, et al. Conventional versus digital impressions for “All-on-Four” restorations. *Int J Oral Maxillofac Implants* 2016;324–30.
  45. Ong AA, Williams FC, Tokarz E, et al. Jaw in a Day: Immediate Dental Rehabilitation during Fibula Reconstruction of the Mandible. *Facial Plast Surg* 2021;37(06):722–7.
  46. Attia S, Wiltfang J, Pons-Kühnemann J, et al. Survival of dental implants placed in vascularised fibula free flaps after jaw reconstruction. *J Cranio-Maxillo-Fac Surg* 2018;46(08):1205–10.
  47. Wu YQ, Huang W, Zhang ZY, et al. Clinical outcome of dental implants placed in fibula-free flaps for orofacial reconstruction. *Chin Med J (Engl)* 2008;121(19):1861–5.
  48. Curi MM, Condezo AFB, Ribeiro KDCB, et al. Long-term success of dental implants in patients with head and neck cancer after radiation therapy. *Int J Oral Maxillofac Surg* 2018;47(6):783–8.
  49. Smith Nobrega A, Santiago JF, de Faria Almeida DA, et al. Irradiated patients and survival rate of dental implants: a systematic review and meta-analysis. *J Prosthet Dent* 2016;116(6):858–66.
  50. Kim DD, Ghali GE. Dental implants in oral cancer reconstruction. *Oral Maxillofac Surg Clin North Am* 2011;23(2):337–45.
  51. Brauner E, Cassoni A, Battisti A, et al. Prosthetic rehabilitation in post-oncological patients: report of two cases. *Ann Stomatol* 2010;1(1):19–25.
  52. Chiapasco M, Biglioli F, Autelitano L, et al. Clinical outcome of dental implants placed in fibula-free flaps used for the reconstruction of maxillo-mandibular defects following ablation for tumors or osteoradionecrosis. *Clin Oral Implants Res* 2006;17(2):220–8.
  53. Kumar VV, Jacob PC, Ebenezer S, et al. Implant supported dental rehabilitation following segmental mandibular reconstruction- quality of life outcomes of a prospective randomized trial. *J Cranio-Maxillo-Fac Surg* 2016;44(7):800–10.
  54. Jacobsen H-C, Wahnschaff F, Trenkle T, et al. Oral rehabilitation with dental implants and quality of life following mandibular reconstruction with free fibular flap. *Clin Oral Invest* 2016;20(1):187–92.
  55. Parbo N, Murra NT, Andersen K, et al. Outcome of partial mandibular reconstruction with fibula grafts and implant-supported prostheses. *Int J Oral Maxillofac Surg* 2013;42(11):1403–8.
  56. Chang YM, Pan YH, Shen YF, et al. Success of dental implants in vascularised fibular osteoseptocutaneous flaps used as onlay grafts after marginal mandibulectomy. *Br J Oral Maxillofac Surg* 2016;54(10):1090–4.
  57. Ciocca L, Corinaldesi G, Marchetti C, et al. Gingival hyperplasia around implants in the maxilla and jaw reconstructed by fibula free flap. *Int J Oral Maxillofac Surg* 2008;37(5):478–80.
  58. De Ceulaer J, Magremanne M, Van Veen A, et al. Squamous cell carcinoma recurrence around dental implants. *J Oral Maxillofac Surg* 2010;68(10):2507–12.
  59. Pinchasov G, Haimov H, Druseikaite M, et al. Oral cancer around dental implants appearing in patients with/without a history of oral or systemic malignancy: a systematic review. *J Oral Maxillofac Res* 2017;8(3):1–10.
  60. Fujiki M, Miyamoto S, Sakuraba M, et al. A comparison of perioperative complications following transfer of fibular and scapular flaps for immediate mandibular reconstruction. *J Plast Reconstr Aesthetic Surg* 2013;66(3):372–5.
  61. Chang YM, Wallace CG, Tsai CY, et al. Dental implant outcome after primary implantation into double-barreled fibula osteoseptocutaneous free flap-reconstructed mandible. *Plast Reconstr Surg* 2011;128(6):1220–8.
  62. He Y, Zhang ZY, Zhu HG, et al. Double-barrel fibula vascularized free flap with dental rehabilitation for mandibular reconstruction. *J Oral Maxillofac Surg* 2011;69(10):2663–9.
  63. Lee JH, Kim MJ, Choi WS, et al. Concomitant reconstruction of mandibular basal and alveolar bone with a free fibular flap. *Int J Oral Maxillofac Surg* 2004;33(2):150–6.
  64. Ernst N, Sachse C, Raguse JD, et al. Changes in peri-implant bone level and effect of potential influential factors on dental implants in irradiated and nonirradiated patients following multimodal therapy due to head and neck cancer: a retrospective study. *J Oral Maxillofac Surg* 2016;74(10):1965–73.
  65. Friedrich RE, Manuel TA, Andreas K. Simulation of scattering effects of irradiation on surroundings

- using the example of titanium dental implants: a Monte Carlo approach. *Anticancer Res* 2010;30(5):1727–30.
66. Korfage A, Raghoobar GM, Slater JJRH, et al. Overdentures on primary mandibular implants in patients with oral cancer: a follow-up study over 14 years. *Br J Oral Maxillofac Surg* 2014;52(9):798–805.
67. Friedrich RE, Todorovic M, Heiland M, et al. Scattering effects of irradiation on surroundings calculated for a small dental implant. *Anticancer Res* 2012;32(5):2043–6.
68. Mancha De La Plata M, Gas LN, Dez PM, et al. Osseointegrated implant rehabilitation of irradiated oral cancer patients. *J Oral Maxillofac Surg* 2012;70(5):1052–63.
69. Zen Filho EV, Tolentino E de S, Santos PSS. Viability of dental implants in head and neck irradiated patients: a systematic review. *Head Neck* 2016;38(S1):E2229–40. Eisele DW.
70. Schoen PJ, Raghoobar GM, Bouma J, et al. Rehabilitation of oral function in head and neck cancer patients after radiotherapy with implant-retained dentures: effects of hyperbaric oxygen therapy. *Oral Oncol* 2007;43(4):379–88.
71. Salinas TJ, Desa VP, Katsnelson A, et al. Clinical evaluation of implants in radiated fibula flaps. *J Oral Maxillofac Surg* 2010;68(3):524–9.
72. Ferrari S, Copelli C, Bianchi B, et al. Rehabilitation with endosseous implants in fibula free-flap mandibular reconstruction: a case series of up to 10 years. *J Cranio-Maxillo-Fac Surg* 2013;41(2):172–8.